

Cybersecurity Training Diary

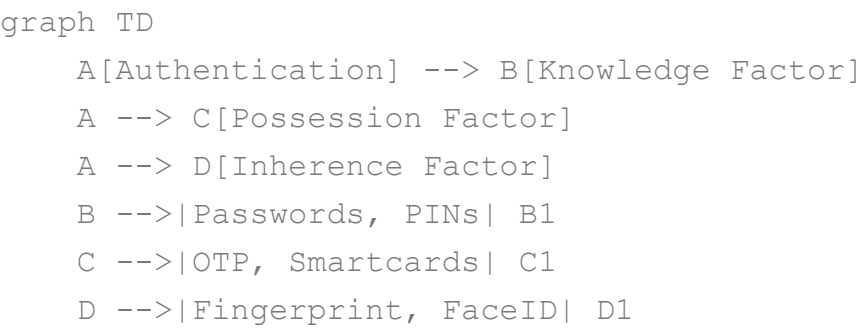
Day 8: Authentication & Authorization

1. Topics Covered

✓ Core Concepts

Term	Definition	Example
Authentication	Verifying identity	Username/password
Authorization	Permission management	Admin vs user roles
Accounting	Tracking access (AAA)	Login audit logs

✓ Authentication Methods



✓ Modern Protocols

Protocol	Purpose	Used In
OAuth 2.0	Delegated access	"Login with Google"
SAML	Enterprise SSO	Company portals
OpenID Connect	Identity layer	Mobile apps

✓ Password Security

- **Hashing:** bcrypt, PBKDF2, Argon2
- **Salting:** Unique random data per password
- **Breach Alert:** HaveIBeenPwned API

2. Key Takeaways

MFA reduces 99.9% of account breaches (Microsoft)

Least privilege principle minimizes damage

Session hijacking is top auth vulnerability

3. Practical Exercises

Exercise 1: Implement bcrypt Hashing (Python)

```
import bcrypt
salt = bcrypt.gensalt()
hashed = bcrypt.hashpw(b"mypassword", salt)
print(hashed)  # Store this in DB
```

Exercise 2: OAuth 2.0 Flow Demo

1. Visit [OAuth Playground](#)
2. Try "Google Calendar API" flow
3. Examine access tokens

4. Questions for Reflection

Why do many companies still use SMS 2FA despite risks?

How would you explain session cookies vs JWTs?

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